Thermal Performance and Radiation Tolerance of Adhesives for the CMS Phase-2 Pixel Barrel Mechanics

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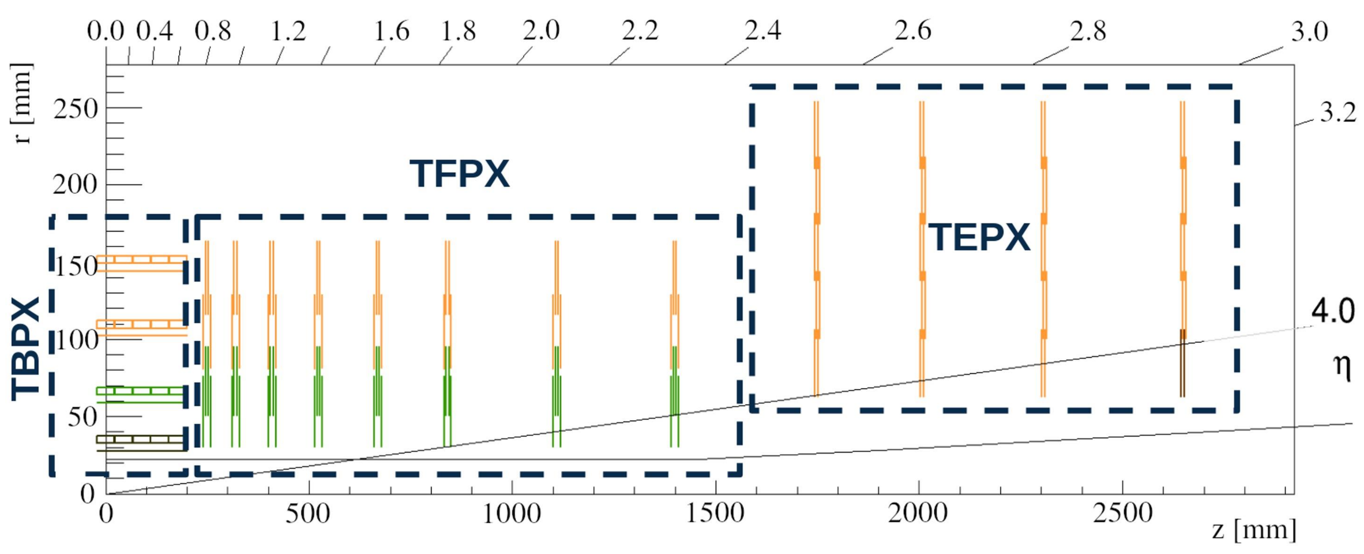
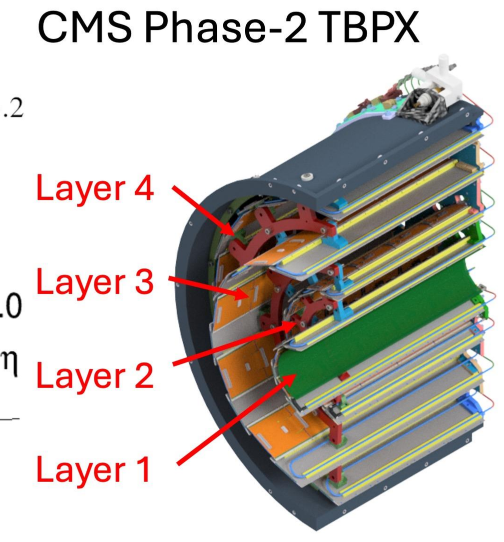
1. Introduction

- The cooling capacity of the CMS Phase-2 pixel barrel mechanics must be significantly higher than that of the first two generations of CMS pixel detectors due to the increased granularity of the pixel readout chip and the corresponding higher power dissipation, as well as much higher expected total irradiation dose. All barrel ladders undergo thorough quality control tests with respect to temperature uniformity and the temperature difference between the CO_2 coolant and the pixel module.
- During prototype testing, numerous adhesives have been tested and evaluated, both on single portions of the structure, and complete layers. Temperature uniformity across all points of the structure and greater thermal efficiency were achieved with less known and provisoly not used glues.
- To verify the radiation tolerance of the selected glues, irradiations with a Co-60 source were performed at RBI Zagreb at doses of 2 MGy, 3 MGy, 6 MGy, and 10 MGy. These doses correspond to the total ionizing dose expected for barrel layers 4, 3, 2, and 1, respectively, at the end of layers lifetime (10 years for layers 2-4 and 5 years for layer 1 of HL-LHC operation).
- The results of the mechanical and thermal characterization of the glues before and after irradiation will be presented, together with the thermal performance of barrel layers constructed using the selected adhesive.

2. CMS Phase-2 Inner Tracker

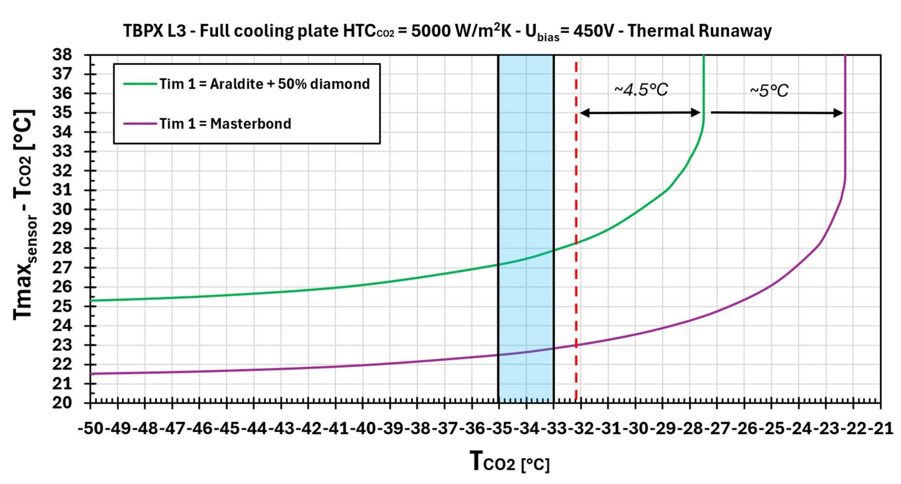




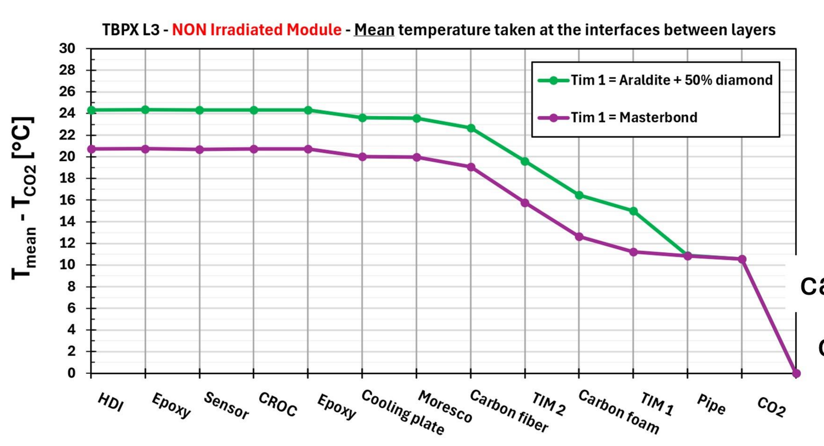
Inner Tracker built from 3892 modules with either 1x2 or 2x2 readout chips
TBPX: Tracker Barrel PiXel, 4 layers 756 modules, ~11kW power
TFPX: Tracker Forward PiXel, 8 small double disks on each side, 1728 modules, ~23kW power
TEPX: Tracker Endcap Pixel Detector, 4 large double disks on each side, 1408 modules, ~21kW
Carbon lightweight mechanics with integrated cooling pipes and two-phase CO_2 cooling system

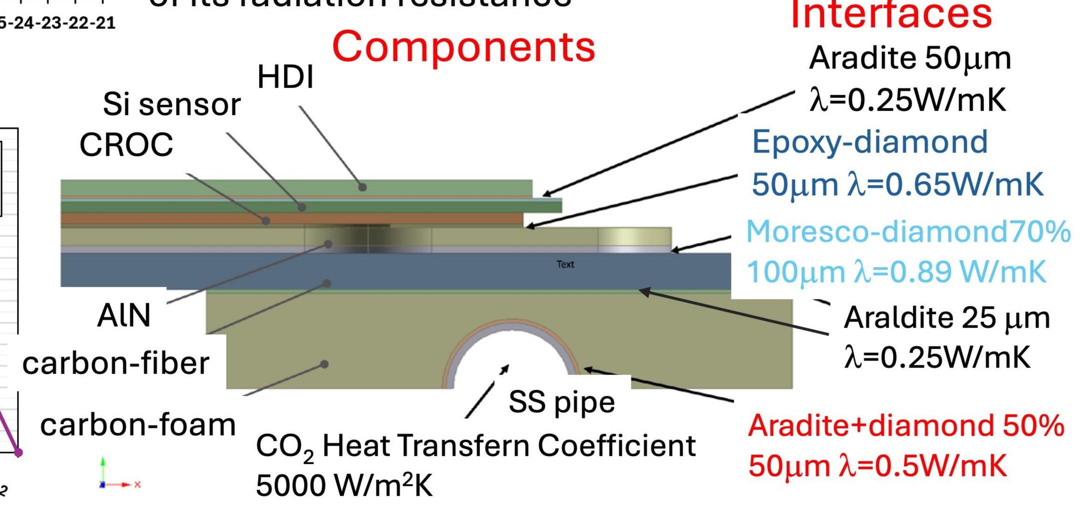
- TBPX pixel modules 4 or 5 per (1/2) ladder with power dissipation of about **12W/module**
- Minimal CO_2 coolant temperature under a pixel module is **-33°C**

3. $\Delta T = Tmax_{Sensor} - T_{CO2}$ simulation of not irradiated modules

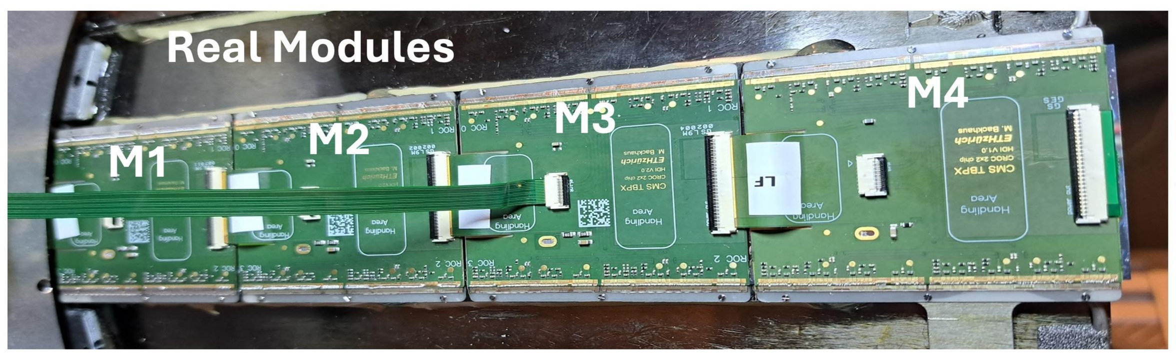


- According to simulations $\Delta T = T_{Module} - T_{CO2}$ for unirradiated module is +25.3°C but measurements show ΔT for some ladders up to +30°C.
- The reason is assigned to low thermal conductivity glue (**Araldite+diamond 50%**) and not perfect contact of glue with cooling pipe and carbon foam
- The new glue **Masterbond EP54TC** as thermal interface material will be used pending confirmation of its radiation resistance

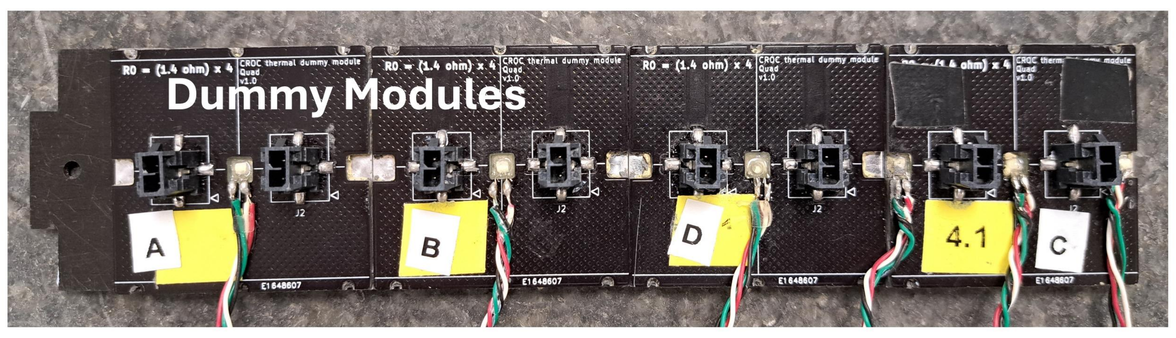




4. Real and dummy modules setups



Real modules: mounted with screws and 100µm of Moresco RG-42R-1 radiation-resistant grease doped with synthetic diamond dust (50% by weight) placed between modules cooling plate and Carbon Fiber ladder in the same way as it will be done in the final detector. T is measured with a built-in temperature sensor.



Dummy modules: PCBs with embedded Cu wire. PCB is glued to Al plate and has a cut in the middle to glue PT1000 to the Al plate for temperature measurements. Modules are mounted on carbon fiber ladder using MULTICOMP PRO Thermal Pads: thickness 1 mm, thermal conductivity 9 W/mK.

Tests performed in climatic chamber $T_{Air} = T_{CO2} + 5°C$
Cooling: MARTA plant with $T_{CO2} = -30$ °C to +15°C

5. Different glue thermal performance

Measurement conditions: $T_{CO2} = +15°C$, $T_{Air} = +20°C$, $P_{Module} = 12W$

Carbon Foam to Cooling Pipe interface: **Araldite + Diamond 50%**

	Power, W	ΔT_M1	ΔT_M2	ΔT_M3	ΔT_M4	<ΔT>
Dummy modules	47.1	27.2	29.9	29.4	27.9	**28.6±1.1**
Real modules	47.1	27.8±1	27.5±0.8	30.0±1.7	-	**28.4±1.2**
Simulation	-	-	-	-	-	**25.3**

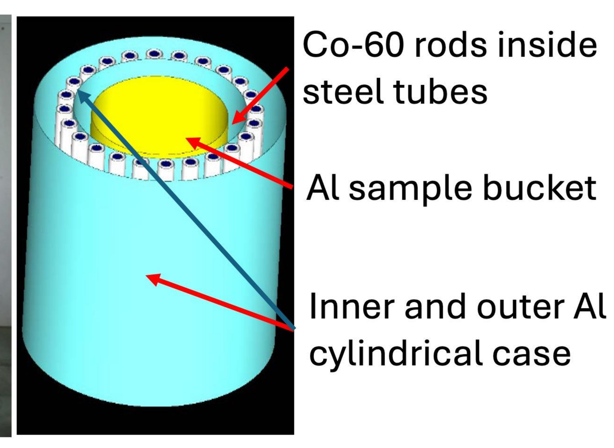
Araldite+Diamond 50%: 50µm λ=0.5W/mK **substituted** with **MasterBond EP54TC:** 50µm λ=6W/mK

Carbon Foam to Cooling Pipe interface: **Masterbond**

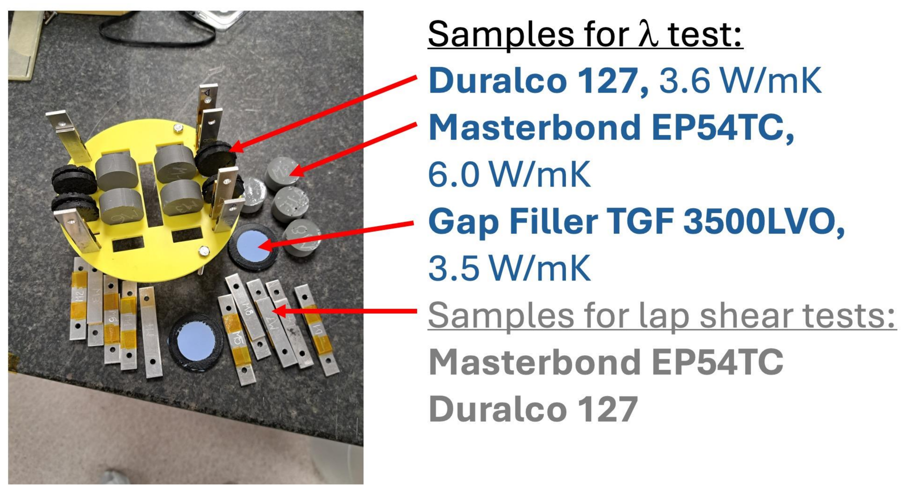
	Power, W	ΔT_M1	ΔT_M2	ΔT_M3	ΔT_M4	ΔT_M5	<ΔT>
Dummy modules	47.1	19.6	20.8	21.9	19.2	18.7	**20.0±1.0**
Simulation	-	-	-	-	-	-	**21.5**

6. Irradiation at Co-60 facility at RBI, Zagreb

- Panoramic Co-60 1.17 1.33 MeV γ–source, 24 rods
- Dose rate: 100 Gy/h in source center, 24/7
- Irradiated volume in source center: cylinder with radius of 20cm and height of 30cm
- Irradiation facility is operated by the Radiation Chemistry and Dosimetry Laboratory of RBI, Zagreb
- Irradiation administrated by CERN: shipping, payment, RP measurements after irradiation

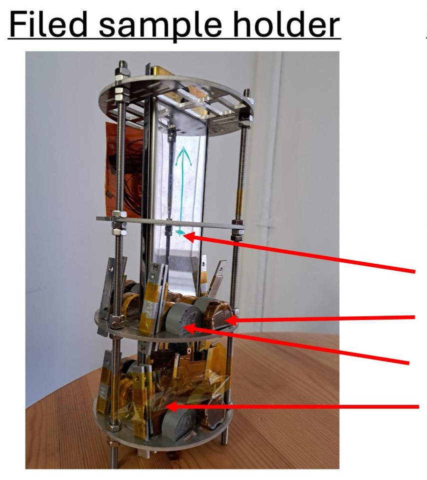


Samples for irradiation



Samples for λ test:
Duralco 127, 3.6 W/mK
Masterbond EP54TC, 6.0 W/mK
Gap Filler TGF 3500LVO, 3.5 W/mK
Samples for lap shear tests:
Masterbond EP54TC
Duralco 127

Filed sample holder



2 samples for each dose:
Layer 4: 2 Mgy (~9 days)
Layer 3: 3 MGy (~13 days)
Layer 2: 6 MGy (~26 days)
Layer 1: 10 MGy (~45 days)
Single modules ladder
Duralco
Masterbond
Gap Filler

7. Thermal & mechanical measurements before/after irradiation

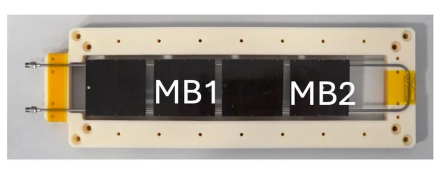
Ladder with cooling pipe glued with Masterbond

T_CO2(C)	16.0	10.7	1.5	-7.7	-16.7	-25.3
0 MGy	21.02	21.82	22.53	22.83	22.98	23.08
3 MGY	21.2		22.21	22.70	23.00	23.10
3 MGY +40 Tcycles*	21.1			22.60	23.00	23.30

No change in T observed after 3 MGy

* Tcycles (-40°C; +30°C)

Single modules ladder



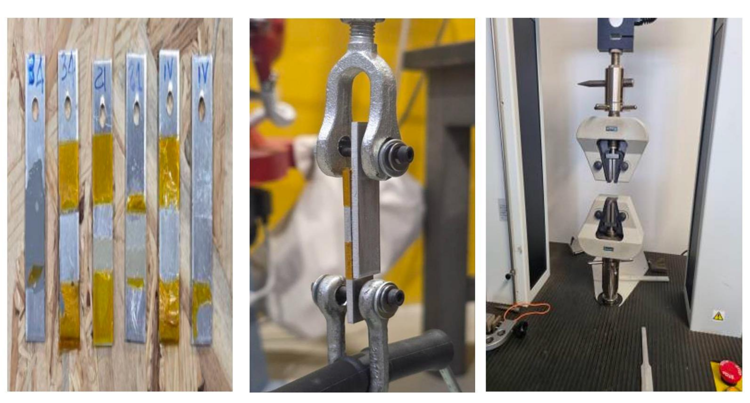
Two MBs gave almost same result

Lap shear test of Al plates glued with Masterbond, curing time 72 hrs, curing temperature 25°C

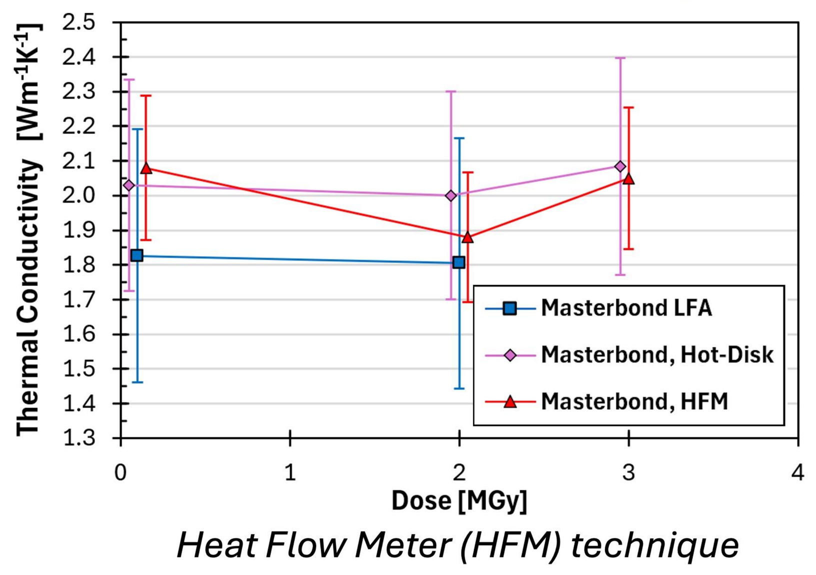
Dose	0 MGy	2 MGy	3 MGy	6 MGy	10 MGy
Breakage Load (kg)	50	35	40	50	40

Post-irradiation measurements up to 10 MGy show no worries on mechanical performance

Samples Sample fistures Tensil tester



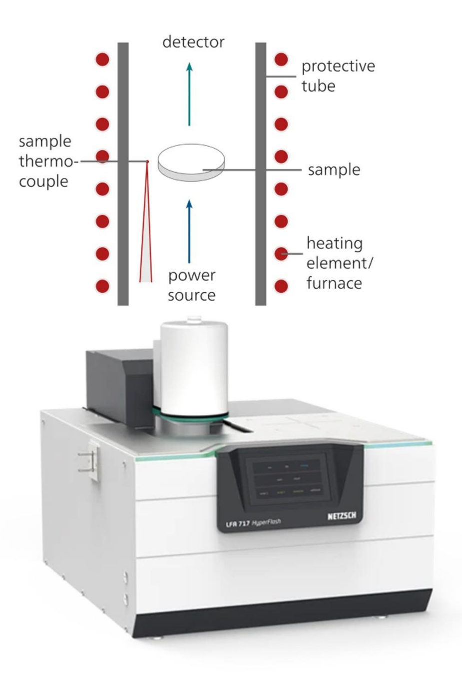
8. Thermal conductivity measurements before/after irradiation



Transient Plane Source (Hot-Disk) technique

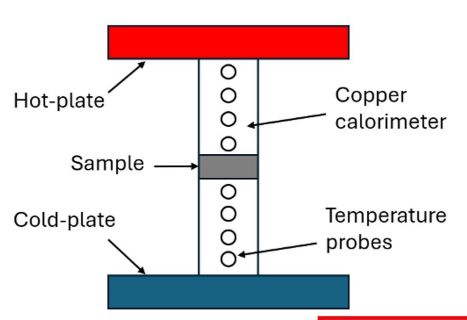
TPS3500 from Hot Disk® Instrument

Light Flash (LFA) technique



LFA 717 Hyperflash® from Netzsch

Heat Flow Meter (HFM) technique



No evidence of thermal conductivity degradation after irradiation!

Final remarks and prospects:

several high thermal conductivity adhesive materials have been irradiated up to 10MGy at the Co-60 irradiation facility at the Rudjer Boskovic Institute in Zagreb, as part of qualification for use in the CMS Inner Tracker. The thermal conductivity and mechanical properties of these adhesives are being evaluated before and after irradiation to assess their radiation hardness and suitability for long-term operation in a high-radiation environment.

Masterbond (highlighted in this poster): demonstrated stable thermal conductivity up to a dose of 3MGy and mechanical stability up to 10MGy. Based on this performance, it can be implemented across all TBPX layers. Furthermore, the L2 and L3 quotas have already been assembled using this new thermal interface.

Duralco (not presented in this poster): not yet been fully tested for thermal and mechanical property retention. If successful, it may be considered for use in the TEPX .

Gap Filler (not presented in this poster): if post-irradiation testing confirms dimensional stability, adhesion strength, and sustained thermal performance, this material could significantly enhance the thermal management efficiency of the IT